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A Breakthrough in Accurate Correlation of RVP to TVP for Crude Oil

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Abstract

This paper presents a rigorous experimental study to establish an accurate and reliable correlation equation for converting Reid Vapor Pressure (RVP) measurements to True Vapor Pressure (TVP) for live crude oils. While RVP is a widely used and standardized measure of volatility, it provides only an empirical snapshot and is often insufficient for modern crude oil streams that contain significant light hydrocarbon components. TVP, by contrast, offers a thermodynamically accurate measure of a crude's volatility under specific operational conditions, which is critical for ensuring tank farm safety, managing evaporative emissions, and complying with stringent environmental regulations. The study was conducted on-site at three major oil and gas processing plants using a portable TVP analyzer to capture live data. The results demonstrate that the widely recognized API MPMS correlation consistently and significantly underestimates the true vapor pressure of these crude oils, confirming its unsuitability for "live" streams. Based on this live dataset, a new, site-specific correlation equation was developed and validated. The new equation, which utilizes a dynamic, empirically determined constant, was shown to provide a high degree of accuracy with minimal percentage error when compared to the direct TVP measurements. This research bridges a critical gap in current industry practice by providing a standardized, validated method to accurately derive TVP from RVP data, thereby enhancing safety, improving emissions control, and ensuring regulatory compliance.

Introduction

True Vapor Pressure (TVP) is a critical physical property that governs the volatility of crude oil and significantly influences its storage, handling, and transportation. As the measure of equilibrium vapor pressure exerted by a liquid at a specified temperature, TVP serves as a direct indicator of a crude's tendency to emit volatile organic compounds (VOCs). Elevated TVP values are particularly problematic during storage in atmospheric tanks, as they increase evaporative losses, raise the risk of flammable atmospheres, and complicate compliance with environmental and safety regulations ([API MPMS Chapter 19.4, 2016](#); [USEPA, 2020](#)).

In the context of floating roof tanks, a widely employed technology to reduce vapor emissions, TVP plays a pivotal role in ensuring structural integrity and operational reliability. High vapor pressure levels can result in excessive emissions through roof seals, deck fittings, and rim vent seals, especially under variable ambient temperatures. These emissions contribute to product loss, air pollution, and non-compliance with

emissions regulations (EEMUA Publication 231, 2012; Smith et al., 2018). Moreover, pressure fluctuations due to high TVP can compromise the mechanical stability of the floating roof, leading to seal degradation, pontoon failure, or even partial submergence of the roof posing serious safety and economic risks (Wang et al., 2019).

The impact of high TVP crude oil is particularly acute in regions with high ambient temperatures, where thermal expansion exacerbates vapor generation. This leads to increased breathing losses and puts additional stress on emission control systems. Studies have established a clear relationship between elevated TVP and higher fugitive emissions from external floating roof tanks, underlining the importance of precise vapor pressure control in tank farm operations (Kumar et al., 2016; EPA, 2020). Additionally, increased vapor accumulation can create potentially explosive atmospheres, especially during filling and draining cycles, demanding rigorous operational oversight.

Given the convergence of environmental, safety, and economic pressures, the accurate measurement and management of crude oil vapor pressure is not merely a technical necessity, but a strategic imperative. The industry relies heavily on Reid Vapor Pressure (RVP) in measuring the vapor pressure due to its simplicity and regulatory familiarity. While RVP is a standardized measurement performed at a fixed temperature of 37.8°C, it provides only a limited, empirical view of a sample's volatility. In contrast, TVP is measured as a function of temperature, offering a comprehensive profile of a crude oil's behavior under various thermal conditions. This paper presents a rigorous experimental study of the widely used RVP method and, crucially, establishes an accurate correlation equation to derive TVP from RVP data. By bridging this critical gap, this research provides a more precise and reliable tool for crude oil volatility management.

Understanding RVP and TVP Measurements

RVP has been widely used for decades as an industry standard measure of petroleum product volatility, valued for its relative simplicity, reproducibility, and long-standing integration into regulatory frameworks and product specifications; particularly for gasoline blending and evaporative emission control (ASTM D323; API MPMS Ch. 19.4, 2016). Defined as the absolute pressure exerted by a liquid at 37.8 °C (100 °F) under a fixed vapor-to-liquid ratio of 4:1, the RVP test provides a convenient empirical indicator of volatility under controlled conditions. Its prevalence is largely historical, stemming from ease of execution and regulatory familiarity, rather than from its ability to replicate real-world storage and transport conditions (Jones & Brown, 2019).

However, in crude oil applications, especially for grades containing significant light ends or exhibiting complex compositional variability, RVP often fails to represent true vapor-liquid equilibrium behaviour. This discrepancy is particularly pronounced at elevated temperatures, in non-ideal mixtures, or in weathered crude oils, leading to underestimation of potential vapor losses and associated safety hazards (API MPMS Ch. 19.4; ASTM D6377). The limitations arise because RVP is measured under a single fixed temperature condition providing no insight into the temperature dependency of vapor pressure, a critical factor in operational environments where storage tanks and pipelines experience wide thermal fluctuations.

TVP in contrast provides a thermodynamically accurate measure of the equilibrium pressure of a liquid vapor system at a specified temperature and can be determined across a range of temperatures and vapor to liquid ratios. Direct TVP determination methods such as the one described in ASTM D6377's simulate operational conditions more effectively by accounting for dissolved gases, flashing behaviour, and compositional interactions (Riazi et al, 2005; ASTM D6377-20). These techniques typically involve placing a crude oil sample into a sealed, evacuated chamber, allowing it to reach thermal and phase equilibrium, and then directly measuring the vapor phase pressure using a precision transducer. This approach captures the dynamic volatility profile of crude oil far more accurately than RVP, particularly for blends where light-end content and compositional changes significantly affect pressure behaviour.

The advantages of direct TVP measurement over traditional RVP are substantial and operationally significant. First, TVP provides data that closely reflects actual storage and transport conditions, enabling precise VOC emission inventories in accordance with environmental guidelines such as USEPA AP-42 and API Technical Report 2569. Second, by accurately identifying the volatility of crude oils, TVP supports safer handling practices, reducing the risk of flammable or explosive atmospheres during tank breathing, loading, and unloading operations (DOE Standards, 2021). Third, TVP measurements can be performed over multiple temperatures, enabling engineers to model vapor pressure behaviour across seasonal or process-related temperature variations.

With regulatory agencies and environmental authorities increasingly favouring TVP-based assessments over RVP for emission permitting and compliance, Petroleum Development Oman (PDO) is transitioning toward direct TVP measurement as a best practice. This shift is driven not only by the need for more accurate engineering data but also by the imperative to reduce environmental impact, enhance operational safety, and maintain product integrity throughout the hydrocarbon supply chain.

RVP to TVP Correlation Equation

While TVP is the most accurate and representative measure of crude oil volatility, historical reliance on RVP has led to the development of correlation equations to convert RVP values to TVP. This approach is often adopted to leverage existing infrastructure with RVP analyzers while attempting to meet the more stringent requirements associated with TVP. However, as this paper will demonstrate, the use of such correlations, particularly for live crude oils, introduces significant uncertainty and can lead to inaccurate assessments of volatility, with serious implications for safety, environmental compliance, and process optimization.

In many oil export terminals, such as at the Petroleum Development Oman (PDO) Terminal, crude oil is stored in large floating-roof tanks designed to operate below a specific TVP threshold to minimize evaporative losses and ensure safety. For example, PDO's tanks are specified to handle crude with a TVP below 86 kPa at a temperature of 45°C. To monitor this parameter, RVP analyzers, based on ASTM D323, are installed at various Main Oil Line (MOL) nodes. Since RVP is measured at a fixed temperature of 37.8°C (100°F) and a specific vapor to liquid ratio, it does not directly correspond to the TVP at the tank's operational temperature of 45°C. Therefore, a correlation is necessary to estimate the TVP from the measured RVP.

In the context of crude oil vapor pressure management, (Campbell, J.M., 2014) developed a nomograph (Fig. 1) that provides an approximate correlation between RVP and TVP. This graphical tool has become a practical reference for converting RVP measurements into equivalent TVP values, particularly in situations where operational or regulatory specifications are expressed in terms of TVP, but routine field measurements are performed using the RVP method. Its application is common in storage and transportation operations where direct TVP measurements are not feasible, offering a convenient, though approximate, means of ensuring compliance with vapor pressure limits.

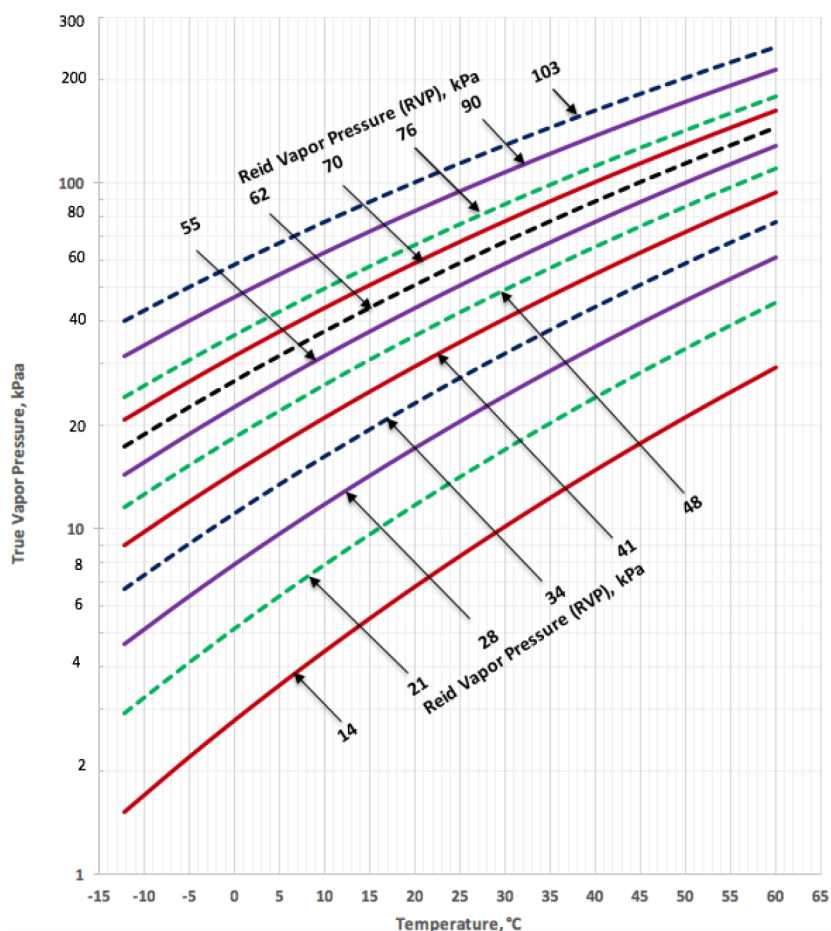


Figure 1—TVP as a function of RVP and temperature for crude oil

The most widely recognized and practiced method for converting RVP to TVP is the API Manual of Petroleum Measurement Standards (MPMS) calculation (API MPMS Chapter 19.4). This method, which often involves complex mathematical equations or nomographs, uses the RVP and the sample's temperature to estimate the TVP. While historically accepted, recent studies and operational experiences have cast doubt on the accuracy of this correlation, particularly for modern crude oil streams.

A recent live measurement study conducted by PDO metering function directly compared correlated TVP values with those obtained from a direct TVP measurement method, specifically the expansion method (VPCR_x) described in ASTM D6377-20. The study found a significant variance between the correlated TVP and the directly measured TVP, highlighting a high degree of uncertainty in the API correlation. This finding is not an isolated case.

The fundamental flaw in many of these correlations, including the API method, lies in the nature of the crude oils used to develop them. A paper titled "Vapor Pressure Measurement Best Practice" by the Canadian Crude Quality Technical Association (CCQTA) explicitly addressed this issue. The CCQTA investigation found that the API correlations were based on "dead" crude oils (CCQTA, 2017). Dead crude oils have been extensively degassed and contain only trace levels of light hydrocarbons such as methane, ethane and propane. Consequently, their vapor pressures are typically less than atmospheric pressure.

In stark contrast, many crude oils transported and processed today are considered "live." Live crudes retain significant quantities of dissolved light hydrocarbons, which can exert substantial vapor pressures, often exceeding atmospheric pressure. Since the API correlations were not developed using these live crudes, they cannot accurately predict the TVP from an RVP measurement for the majority of modern

crude oil streams. This mismatch between the correlation's foundational data and the characteristics of contemporary crude oils is the primary source of the observed uncertainty and inaccuracy (Riazi et al, 2005).

This paper addresses the critical need for a more accurate RVP-to-TVP conversion method. We present a rigorous experimental study that systematically investigates the relationship between RVP and TVP for a range of "live" crude oils. The study utilizes both the standard ASTM D323 RVP method and the direct TVP measurement method (ASTM D6377-20) to generate a robust dataset.

The core of this research is the establishment of a new, scientifically derived correlation equation. The proposed correlation is developed through a comprehensive statistical analysis of the experimental data, taking into account key parameters such as crude oil composition and temperature. By moving beyond the outdated assumptions of the API correlation, our new equation provides a more precise and reliable tool for crude oil volatility management. This research bridges a critical gap in current industry practice, offering a standardized, validated method to accurately derive TVP from RVP data, thereby enhancing safety, improving emissions control, and ensuring regulatory compliance.

Experimental Methodology for RVP-TVP Correlation

To accurately establish a correlation between RVP and TVP, a rigorous experimental methodology was developed and implemented. The study was conducted through live, on-site measurements of crude oil samples from three distinct oil and gas processing plants (hereafter referred to as plants A, B and C). This multi-site approach ensured a diverse range of crude oil compositions and operating conditions were sampled, increasing the robustness of the resulting data.

Instrumentation and Measurement

The experimental setup involved two key types of instruments: online RVP analyzers and a laboratory portable TVP analyzer.

- **Online RVP Analyzers:** RVP data was continuously collected using online process analyzers, which operate based on the ASTM D323 standard. At plant A, two analyzers were positioned at the discharge of the two condensate stabilization trains, with a third analyzer on the common discharge header, providing multiple data points for a single facility. Plants B and C each had one online RVP analyzer. All online analyzers were calibrated and stabilized prior to the start of the experiment to ensure data integrity.
- **Portable TVP Analyzer: Lab portable TVP analyzer** was used for all TVP measurements. This analyzer employs the expansion method (VPCR_x) as described in ASTM D6377. Its key features made it ideal for this study:
 - **Portability:** Rugged and compact design, the analyzer allows for rapid, on-site testing in a mobile lab, avoiding the uncertainties and potential losses associated with transporting samples
 - **Versatility:** Can measure crude TVP across a wide range of temperatures, from 20°C to 60°C, allowing for the precise replication of TVP at various operational temperatures, including the target temperature of 45°C. It also determines TVP over a pressure range of 0-1000kPa.
 - **Variable Vapor-to-Liquid (V/L) Ratio:** The analyzer's sample conditioning system facilitates the direct transfer of a pressurized crude oil sample into the measuring chamber. A single expansion of an integrated piston determines the vapor pressure against a vacuum. Subsequently, a multi-point expansion sequence is performed at various V/L ratios. A curve fit extrapolation to a V/L ratio of 0/1 is then conducted to determine the true TVP, which accounts for the dissolved gases in the sample (Fig. 2). This comprehensive approach provides a more accurate representation of the crude's volatility than a single-point measurement.

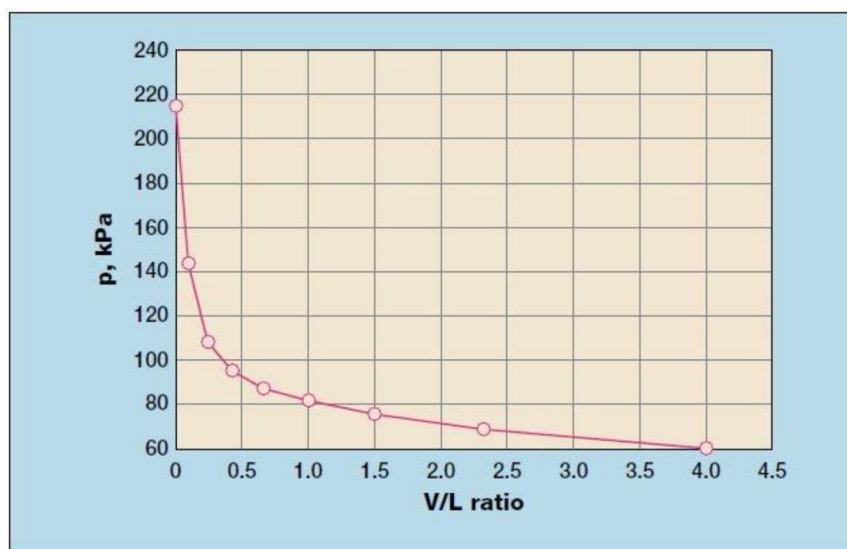


Figure 2—Vapor pressure of crude oil at different vapor-liquid ratios at 37.8 °C (illustrative figure)

Data Collection Procedure

The experimental program was conducted over several days to capture the variability in crude oil properties and plant operating scenarios. To ensure the samples were representative, the portable Lap TVP analyzer was brought to each site. A crucial aspect of this methodology was the direct, on-site measurement of TVP, which minimized the potential for light end flashing and composition changes that can occur during sample transport.

For each experimental run, the following steps were performed:

1. All online RVP and portable TVP analyzers were calibrated and verified.
2. At each plant, multiple crude oil samples were taken at different times and under varying plant operating conditions.
3. Each sample's RVP was recorded from the online analyzer.
4. Simultaneously at the beginning of the RVP measuring cycle, a sub-sample was transferred to the portable TVP analyser to measure its TVP. This simultaneous measurement of RVP and TVP on the same sample at the same time was critical for establishing a valid correlation.

By systematically collecting paired RVP and TVP data points across a broad spectrum of live crude oils, this methodology provides a robust foundation for developing an accurate correlation equation that is relevant to modern petroleum operations.

Results

The experimental methodology yielded a comprehensive dataset of paired RVP and TVP measurements for crude oil samples collected from the three designated production plants. The data, summarized in [Table-1](#), provides a detailed comparison between the standard RVP measurements, the TVP values calculated using the API correlation, and the direct TVP measurements obtained with the portable TVP analyzer at various vapor-to-liquid (V/L) ratios.

Table 1—Experimental results of RVP and TVP measurements

Train-Sample No:	Online RVP (PSI)	Online RVP (Kpa)	Calc. TVP API formula @ 45°C	Lab analyzer TVP at V/L ratio (Kpa) @ 45°C:					
				0.1	0.25	0.5	1	4	TVP at 0
A TR2-1	5.22	35.99	53.36	171	131.46	104.89	83.25	61.13	224.95
A TR2-2	5.3	36.54	54.31	177.57	136.2	108.53	85.99	63.02	233.26
A TR2-3	5.36	36.96	55.02	152.1	120.68	99.47	81.92	63.22	191.47
A TR2-4	6.03	41.58	63.04	175.96	138.14	112.43	91.18	68.59	223.62
A TR2-5	6.09	41.99	63.77	182.77	143.25	116.51	94.61	71.39	233.1
A TR2-6	7.04	48.54	75.40	174.63	139.83	116.39	97.11	75.99	216.98
A TR2-7	5.58	38.47	57.64	163.86	131.57	109.98	92.19	72.96	203.72
A COM-1	6.17	42.54	64.74	205.98	156.84	123.92	97.42	70.6	273.47
A COM-2	6.36	43.85	67.05	192.89	148.15	118.33	94.27	69.64	253.3
A COM-3	6.31	43.51	66.44	206.2	156.28	122.85	95.93	68.76	274.94
A COM-4	5.95	41.03	62.08	184.61	141.66	112.82	89.58	65.83	242.79
A TR1-1	5.57	38.41	57.52	140.36	116.94	98.76	82.38	61.87	163.62
A TR1-2	5.4	37.23	55.49	152.78	121.44	99.7	81.27	60.49	188.76
A TR1-3	5.04	34.75	51.24	180.97	137.89	109.08	85.74	61.5	237.84
A TR1-4	5.33	36.75	54.66	187.59	144.89	115	90.11	63.74	241.46
B -1	8.63	59.50	95.40	174.69	134.38	107.7	86.33	64.17	228.57
B -2	9.78	67.4	110.19	209.98	172.24	145.64	123.17	98.18	254.3
B -3	10.34	71.3	117.59	207.49	171.67	146.52	125.23	101.42	249.17
B -4	9.06	62.5	100.98	193	159.69	136.46	116.75	94.97	232.21
B -5	9.12	62.9	101.73	197.01	162	137.56	116.99	94.52	239.12
B -6	8.64	59.6	95.59	189.68	155.62	132.04	112.18	90.69	231.05
B -7	8.95	61.7	99.49	187.97	155.8	133.34	114.39	93.56	226.27
B -8	9.65	66.5	108.49	188.55	156.33	133.76	114.7	93.73	226.86
C -1	12.77	88.05	150.06	270.07	219.27	180.09	147.38	106.26	329.93
C -2	12.77	88.05	150.06	139.14	134.05	128.51	122.03	106.74	142.02
C -3	6.96	47.99	74.41	127.92	116.79	107.88	99.74	86.91	136.39
C -4	7.29	50.26	78.50	113.37	105.41	98.98	93	82.95	118.96
C -5	9.19	63.37	102.60	131.35	120.82	112.33	104.42	91.06	138.7
C -6	6.02	41.51	62.92	113.88	103.35	95.24	88.06	77.34	122.47
C -7	7.34	50.61	79.12	123.54	110.78	100.93	92.28	79.63	134.24
C -8	9.66	66.61	108.68	134.55	121.57	110.88	101.15	85.8	144.55
C -9	10.92	75.29	125.23	149.36	136.9	126.5	115.48	96.59	157.58
C -10	8.16	56.26	89.43	114.78	108.04	102.4	96.53	85.24	118.73
C -11	6.98	48.13	74.66	110.35	103.34	97.3	91.3	80.52	114.93

The first observation from the data is the clear and consistent discrepancy between the calculated TVP values from the API formula and the directly measured TVP at a V/L ratio of 0/1. For instance, sample TR1-1 from the plant A, with an RVP of 35.99kPa (5.22 PSI), has a calculated TVP of 54.31kPa using the API correlation. However, the direct measurement from the lab analyzer shows a TVP of 224.95kPa. This difference is not an isolated event; a similar pattern is evident across all samples, including those from plants B and C.

- Significant Underestimation:** The API correlation consistently and drastically underestimates the true vapor pressure (Fig. 3). For many samples, the calculated TVP is only a small fraction of the actual measured value. For example, sample C -1 has an RVP of 88.05kPa (12.77 PSI), and the API formula predicts a TVP of 150.06kPa. In stark contrast, the portable TVP analyzer records a TVP of 329.93kPa. This finding strongly supports the hypothesis that the API correlation, based on "dead" crudes, is not suitable for the "live" crudes being processed at these facilities.

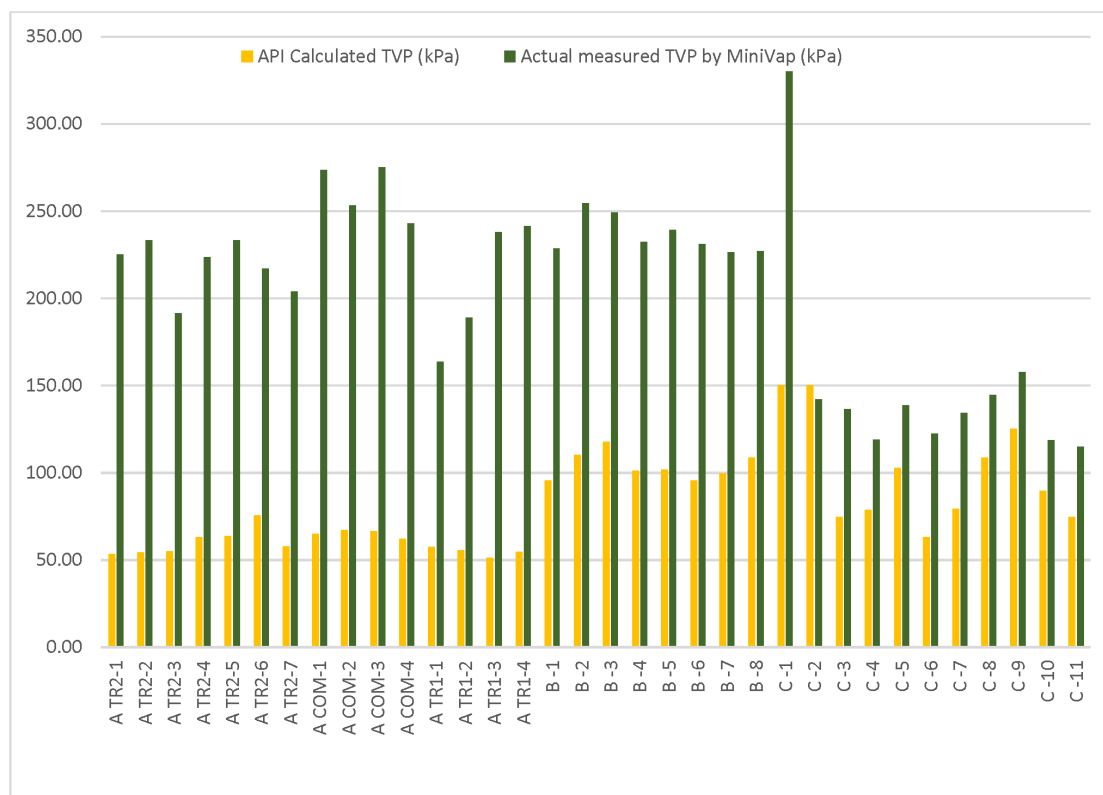


Figure 3—Comparison between the TVP values calculated using the API correlation, and the direct TVP measurements

- Impact of Light Ends:** The significant underestimation by the API method is a direct consequence of the presence of light ends in the crude oil. The online RVP analyzers, while consistent, do not fully capture the influence of these dissolved gases. The TVP expansion method, by contrast, accounts for the flashing behaviour of these volatile components, leading to a much higher and more accurate TVP value.

The data from the portable TVP analyzer, which measures TVP at different V/L ratios (0.1, 0.25, 0.5, 1, and 4), provides a comprehensive volatility profile for each crude sample. This multi-point measurement is crucial for understanding how the crude oil's vapor pressure changes as it is degassed.

- V/L Ratio Dependency:** As expected, the TVP value is shown to be highly dependent on the V/L ratio. For a given sample, the measured pressure decreases as the V/L ratio increases. This is because at higher V/L ratios, the same amount of vaporized light ends is distributed over a larger vapor volume, resulting in a lower measured pressure.
- Extrapolation to 0/1:** The TVP at a V/L ratio of 0/1, which is the final extrapolated value, represents the true equilibrium vapor pressure. This value is consistently the highest recorded pressure and serves as the benchmark for comparison against the RVP and API correlation results. This highlights the importance of the expansion method's capability to extrapolate to a true equilibrium state, a feature entirely absent in the RVP and API correlation methods.

The results from this experimental study conclusively demonstrate the unreliability of using the API correlation for converting RVP to TVP for the live crude oils sampled. The API formula consistently produces values that are significantly lower than the actual measured TVP, a discrepancy that could lead to substantial underestimations of tank emissions and potential safety risks. The direct TVP measurements from the direct TVP measuring analyzer not only provide a more accurate and reliable value but also offer a detailed volatility profile by accounting for the sample's behaviour across a range of V/L ratios. This dataset

provides a strong empirical foundation for the development of a new, more accurate correlation equation that is specifically tailored to the characteristics of modern live crude oils.

Development of a Site-Specific RVP-TVP Correlation

The significant discrepancies observed between the API correlation and the live TVP measurements necessitated the development of a new, more accurate correlation equation. The goal of this effort was to create a robust model that could reliably convert RVP data from existing online analyzers into a representative TVP value for the "live" crude oils at each specific plant.

Instead of creating a new model from scratch, the development process began by leveraging the established structure of the API equation. A detailed analysis of the API formula revealed that its primary limitation for "live" crudes was its reliance on fixed, non-adjustable constants. Through an iterative regression analysis using the experimental RVP and TVP data, it was concluded that a key constant within the API equation needed to be dynamic and site-specific to accurately reflect variations in crude oil composition, particularly the volume fraction of light end components.

This analysis led to the development of a new correlation equation, where a crucial constant, represented as **A1**, is treated as a dynamic parameter. The new, optimized correlation equation is presented as follows:

$$TVP(kPa) = \left[\left(\frac{2799}{T(^{\circ}F) + 459.6} \right) - 2.227 \right] \log_{10} RVP(PSI) - \left(\frac{7261}{T(^{\circ}F) + 459.6} \right) + A1 \bigg] * 6.895$$

Equation 1 -
Accurate RVP to
TVP Correlation
equation

Here, T is the temperature in degrees Fahrenheit, RVP is the Reid Vapor Pressure in pounds per square inch (PSI), and **A1** is a site-specific, experimentally determined constant. The final term of 6.895 is a conversion factor to express the final TVP in kilopascals (kPa).

The constant **A1** was empirically determined for each of the three sites (A, B and C) by fitting the experimental RVP and TVP data to the new correlation equation. The value of **A1** was found to be dependent on the crude oil's composition and also varied slightly with the vapor-to-liquid (V/L) ratio. The optimized values for A1 at a temperature of 45°C across different V/L ratios are summarized in [table- 2](#) below:

Table 2—A1 constants at different V/L Ratios for plants A, B and C

TVP @ 45C	A	B	C
V/L=0.02	14.13	13.66	13.32
V/L=0.1	13.87	13.46	13.22
V/L=0.25	13.62	13.26	13.14
V/L=0.5	13.42	13.10	13.06
V/L=1	13.21	12.93	12.99
V/L=4	12.93	12.69	12.86

The variation of **A1** across the sites and with the V/L ratio underscores the importance of a dynamic constant. The values range from approximately 12.86 to 14.13, depending on the specific crude composition at each plant. This confirms that the volume fraction and type of light end components are the primary factors influencing the correlation between RVP and TVP, and that a single, universal constant is insufficient for accurate prediction.

Table-3 provides a detailed quantitative validation of the new RVP-TVP correlation equation. It presents a side-by-side comparison of the TVP values calculated using the new model against the actual TVP measurements obtained from the laboratory portable analyzer. Crucially, the table also quantifies the percentage error between the developed equation and the lab measurements, providing a robust statistical basis for evaluating the new model's performance.

1. **High Accuracy at TVP (V/L = 0/1):** The "Error between Developed new Eq. and Lab TVP at V/L= 0/1" column demonstrates the remarkable accuracy of the new correlation equation. For the vast majority of samples, the percentage error is in the single digits, indicating a very close fit between the predicted TVP and the actual measured TVP. This is a significant improvement over the API correlation, which, as previously shown, often yielded errors exceeding 100%. The new equation's ability to accurately predict the true equilibrium vapor pressure at a V/L ratio of 0/1 is the most critical finding of this study. Fig.-4 visually represents this key finding. The red line, which plots the TVP values from the New Equation, shows a remarkable agreement with the Actual Measured TVP (green bars). The red line closely follows the peaks and troughs of the green bars, indicating that the new equation effectively captures the true vapor pressure profile of the crude oil samples. The new correlation equation successfully accounts for the variability in crude composition and accurately predicts TVP across a wide range of samples from all three sites.

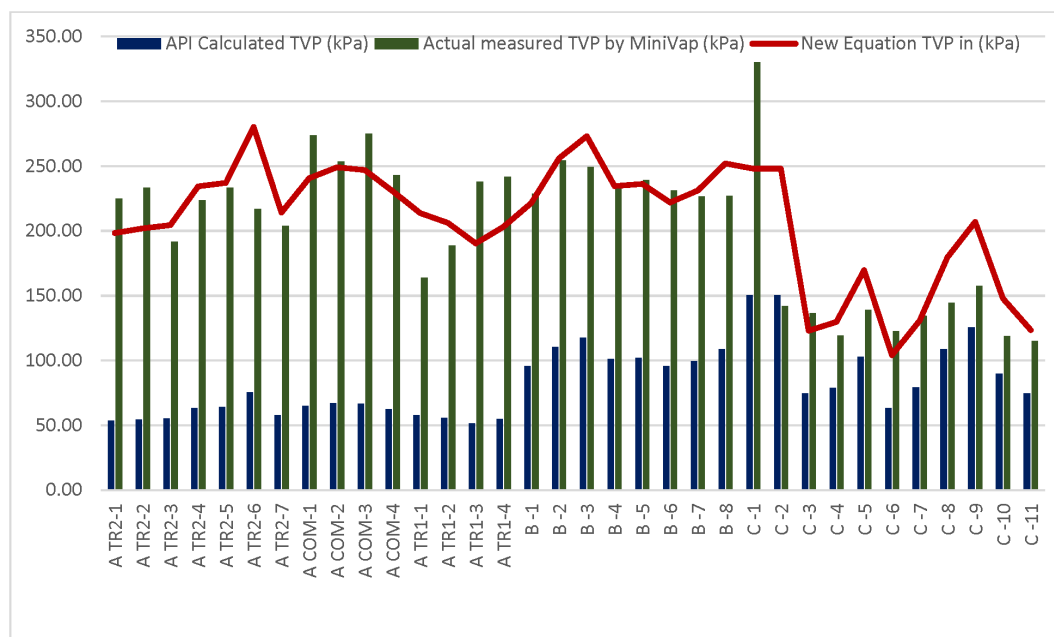


Figure 4—Comparative analysis of the New RVP-TVP correlation Equation Vs API calc. TVP and directly Measured TVP for various crude oil samples

- Consistent Performance Across V/L Ratios:** The analysis extends beyond the extrapolated TVP at a V/L ratio of 0/1. The table shows that the new correlation equation also maintains a high degree of accuracy when predicting TVP at various other V/L ratios (0.1, 0.25, 0.5, 1, and 4). While the errors fluctuate slightly, they remain consistently low across the board. This is particularly important for applications where TVP at a specific, non-zero V/L ratio is required, such as for simulating tank breathing or vapor recovery unit performance.
- Robustness Across Different Crude Sources:** The low percentage errors are consistently observed for samples from all three plants (A, B and C). This confirms that the site-specific constants (**A1**) effectively adapt the correlation equation to the unique crude compositions at each location. The model is not just accurate for one type of crude but is a reliable and robust tool for all three production streams investigated.

The data in the table provides conclusive quantitative evidence that the newly developed RVP-TVP correlation equation is a highly accurate and reliable tool for predicting the true vapor pressure of live crude oils. By using site-specific constants derived from direct measurements, the model overcomes the inherent limitations of the outdated API correlation. The consistent low percentage error across a range of samples and V/L ratios validates the methodology and confirms that the new equation can be confidently implemented for real-time TVP monitoring, thereby enhancing operational safety, improving environmental compliance, and optimizing crude oil management.

Conclusions

The data from this study provide conclusive quantitative evidence that the newly developed RVP-TVP correlation equation is a highly accurate and reliable tool for predicting the true vapor pressure of live crude oils. By using site-specific constants derived from direct measurements, the new model overcomes the inherent limitations of the API correlation, which was shown to consistently produce values significantly lower than actual TVP.

The key conclusions drawn from this research are:

- **Superior Accuracy:** The new correlation equation demonstrates remarkable accuracy, with a low percentage error in predicting TVP values across a range of samples and vapor-to-liquid ratios. This provides a robust and reliable alternative to traditional correlations.
- **Enhanced Operational Safety and Compliance:** The implementation of this new correlation allows facilities to leverage existing online RVP analyzers to obtain accurate, real-time TVP data. This enhanced visibility is crucial for making informed operational decisions, ensuring crude oil is maintained within safe vapor pressure limits for storage and transportation, and aligning with regulatory requirements.
- **Robustness and Scalability:** The site-specific constants effectively adapt the correlation to the unique crude compositions at each plant, demonstrating the model's robustness. This methodology provides a scalable framework for other facilities to develop their own accurate RVP-to-TVP correlations, thereby improving volatility management across the industry.

In summary, this research not only validates the need for a new approach to RVP-to-TVP conversion but also provides a proven, accurate, and practical solution that can be confidently implemented to enhance operational safety, improve environmental performance, and optimize crude oil management.

Reference

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